

PROJECT DOCUMENTATION

Simulating the 2026 FIFA World Cup

Three engines, one ablation, and the number I actually trust.

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A complete record of the experiments: a calibrated statistical Monte Carlo, a fully agentic simulator with 1,248 LLM player agents, the ablation study that separated tactics from noise, and the pure prediction engine that produced the final win percentages.

Contents

1. Why I built this
2. The data
3. Engine one: calibrated statistical Monte Carlo
4. Engine two: the agentic simulator
5. Results from both engines
6. The ablation study
7. The prediction engine and the final answer
8. What I learned
9. Limitations and next steps
10. Artifact index

1. Why I built this

I started with a simple question that turned out to have a sharp edge: can large language model agents make a football simulation smarter without breaking its physics? Anyone can wire a chatbot to a dice roll. The hard part is letting agents make real decisions while the match still behaves like football, and then proving, with numbers, whether those decisions matter.

Over the course of the project the goal split into two honest halves. The first half was a research question about agents: do LLM decisions change who wins a simulated tournament? The second half was the practical question everyone actually asks: who is going to win the 2026 World Cup, and with what probability? By the end I had an answer to both, and the two answers point in different directions, which is the most interesting result of the whole study.

2. The data

Everything starts with real data, because a simulator is only as good as what it eats.

Real squads

I parsed FIFA's official squad list PDF, 48 pages with one squad per page, into a CSV of 1,248 real players: name, shirt number, position, age, club, caps, and international goals. Each player then gets a rating built from real signals: club strength, caps, an age curve that peaks around 27, and team context.

Engineering note. The PDF repeats shirt names inside full names, so my first parse produced players like "Vinicius Vinicius Junior" and a Spanish midfielder called "Rodrigo Rodri". One accent-insensitive deduplication rule (drop the given name when it equals or prefixes a surname token) fixed 170 names.

Team features

Team strength comes from a weighted blend of World Football Elo ratings, squad market value, SPI style ratings, and FIFA points. The 2026 field is stored as an offline snapshot so every run is reproducible. The same feature pipeline covers the 2010 through 2022 World Cups, which is what makes honest backtesting possible.

A note on synthetic versus real players

A question I got early: did the original statistical engine know about individual players, or just teams? The honest answer is yes and no. It always had a full per-player layer: ratings, fatigue, and lineup cohesion feeding into team strength. But the identities were synthetic placeholders shaped like a national squad. The real names, clubs, and caps only entered the project with the agentic engine and the parsed squad lists.

3. Engine one: calibrated statistical Monte Carlo

The first engine is a classical Monte Carlo simulator with one modern twist. The core is a minute-by-minute Poisson goal process: each minute, each team generates scoring chances at a rate driven by attack and defense cohesion, fatigue, and tactical modifiers. Group standings, the third-place matrix, and the official 48-team bracket (twelve groups, a round of 32, then knockouts) are all implemented exactly as FIFA plays them, and matches that have already been played in the real tournament are pinned to their real scorelines.

The twist is a single LLM tactical brain per side. Before each match, gpt-4o-mini reads a scouting brief and picks a mentality and pressing level, which become bounded tactical modifiers on the goal process. This engine ran 100 full tournaments, roughly 20,000 LLM calls in total.

Validation

Before I trusted it on 2026, I backtested the engine on the 2010, 2014, 2018, and 2022 World Cups using only information available before each tournament. The calibration target was the real World Cup scoring average of about 2.64 goals per match, and the engine held it.

Engineering note. After thousands of API calls the OpenAI endpoint started returning "431 Request headers are too large". The SDK's HTTP client had been silently accumulating Cloudflare cookies on every response. The fix was a cookie-blocking jar plus a three-strike retry breaker. Separately, my config file had Windows line endings, and shell-sourcing the API key left a trailing carriage return that produced an illegal HTTP header. Small things break large runs.

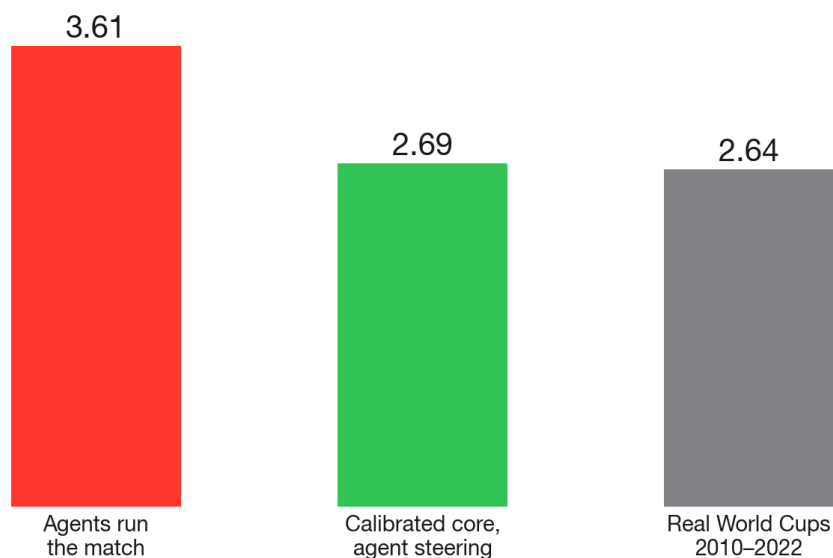
4. Engine two: the agentic simulator

The second engine is the one the project is really about. Every squad becomes an organization of agents: a manager who writes game plans and reacts at half time, outfield players who decide how to attack, and goalkeepers who decide how to defend. All of them are played by qwen2.5 14B, an open model running on one rented RTX 4090 for about fifty cents an hour.

Version one failed, and the failure was the lesson

My first version let the agents run the match directly: their choices were the simulation. It produced 3.61 goals per match against a real World Cup average of 2.64. Ninety-five percent of goals came from passing moves, because the model simply likes passing and my rules rewarded it, and the top scorer finished on eleven goals. The simulation was not simulating football. It was simulating the model's personality.

Goals per simulated match — nearly 7,000 matches



Goals per simulated match. Letting agents run the match broke calibration; the rebuilt architecture restored it.

The fix: calibrated core, agentic steering

The rebuilt architecture keeps the validated statistical engine as the physics and lets agents steer it at exactly three points, nowhere else:

- **Pre-match.** The manager writes a plan in plain language. It is compiled into tactical modifiers clamped to roughly plus or minus fifteen percent. An agent can tilt a match. It cannot rewrite it.
- **Every goal.** Each engine-generated scoring moment becomes a duel. The attacker chooses shoot, pass, or dribble; the keeper chooses hold the line, rush out, or narrow the angle. The keeper can deny the chance outright, and about one chance in ten dies here. The engine

compensates its expected-goals math for the expected denial rate, so the average stays honest while individual matches gain texture.

- **Half time.** The manager reads the actual scoreline and makes real substitutions, and team chemistry is recomputed over the new eleven.

Calibration at scale

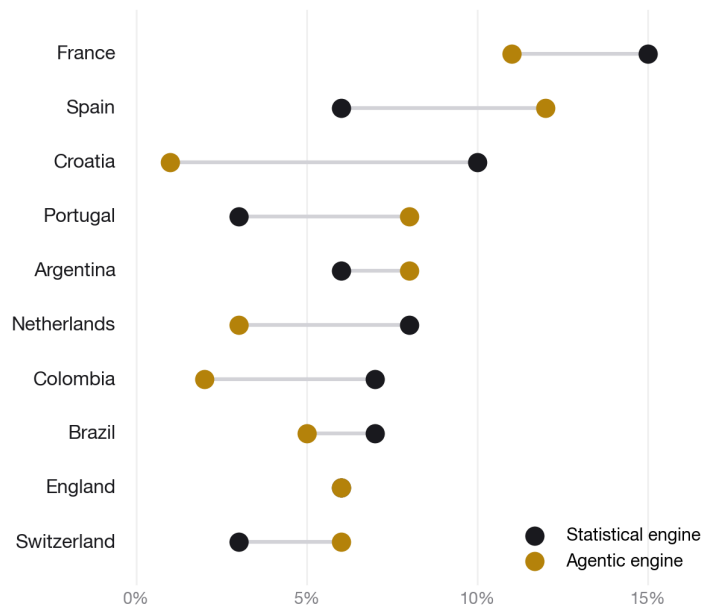
Across nearly 7,000 simulated matches the rebuilt engine averaged 2.69 goals per match against the real 2.64, and the keeper denial rate came out at 10.1 percent against a 10 percent design target. The LLM accounting was clean: over 100,000 calls, zero transport failures, and 0.5 percent of responses falling back to safe defaults, counted rather than silently swallowed.

5. Results from both engines

Each engine played 100 full tournaments. They agree on the favorites and disagree on the order.

Rank	Statistical engine	Win %	Agentic engine	Win %
1	France	15	Spain	12
2	Croatia	10	France	11
3	Netherlands	8	Argentina	8
4			Portugal	8
5			England / Switzerland	6

Chance of winning the 2026 World Cup — 100 tournaments per engine

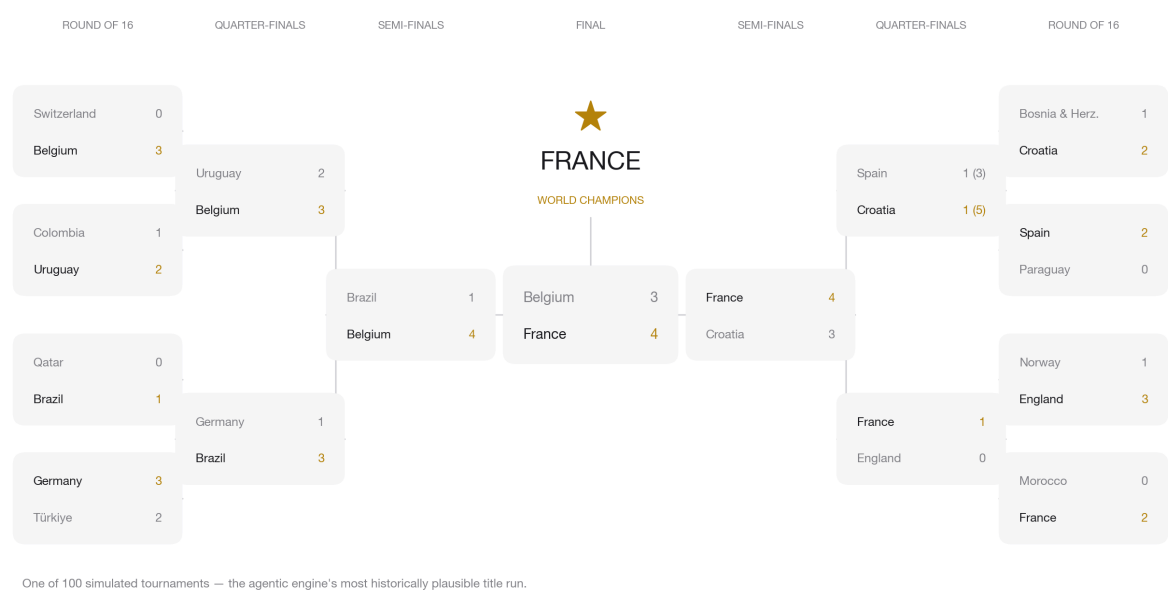


Title odds under both engines, 100 tournaments each.

The agentic field was strikingly flat: 25 different champions in 100 runs. That flatness is the observation that forced the ablation study in section 6.

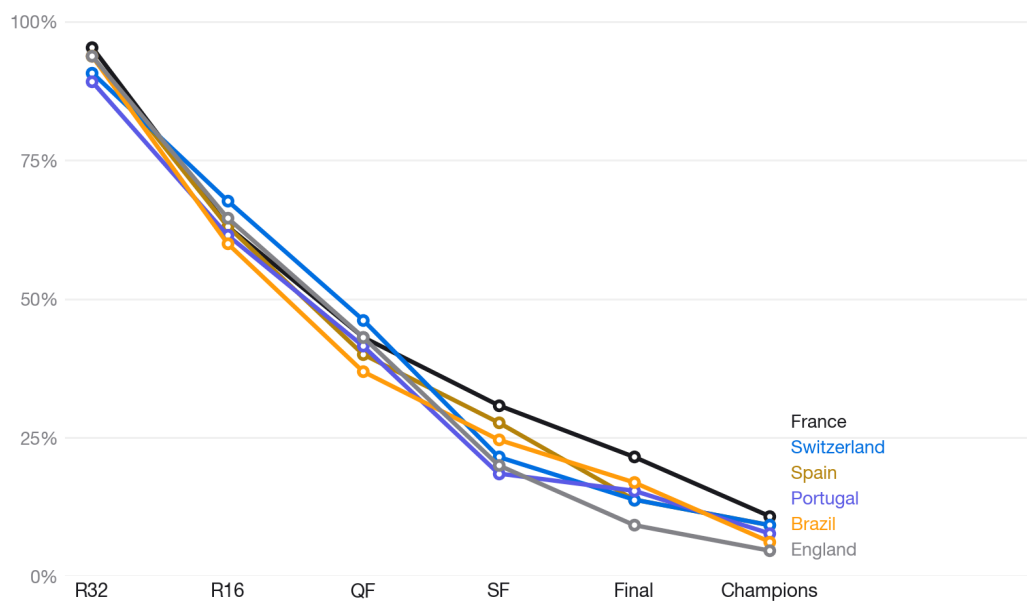
One tournament, end to end

Numbers are abstract, so I also picked the single most historically plausible title run the agentic engine produced and drew its full bracket. What it invented on its own is remarkable: Belgium knocking out Brazil, which is 2018; Croatia winning a penalty shootout, which is practically a law of nature; and France winning a 4-3 final. A 14 billion parameter model making thousands of small decisions re-derived the emotional arc of the 2018 World Cup.



The most historically plausible simulated title run, round of 16 through the final.

How far the favourites survive — agentic engine, stage by stage



Stage-by-stage survival for the favorites. The quarter-finals are the cliff, exactly where real World Cups break hearts.

6. The ablation study

Twenty-five champions in 100 runs could mean two very different things. Either the agents create genuine tactical chaos, or the spread is just sampling noise wearing a tactics costume. There is exactly one experiment that settles it: remove the brains and see if anything changes.

Design

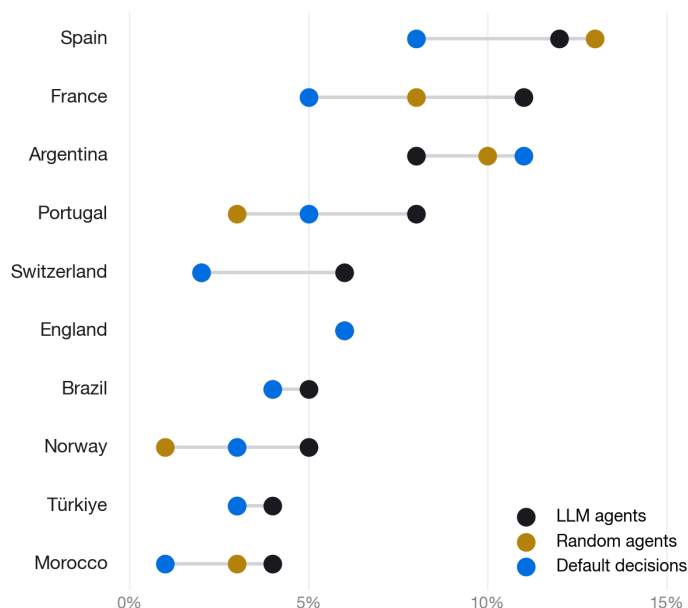
I added a random-agents mode to the engine. It keeps the identical pipeline, the identical decision points, and the identical action space, but every choice is drawn uniformly at random. A manager picks a random mentality and random substitutions; a striker flips a three-sided coin between shoot, pass, and dribble; a keeper does the same with its three options. I also ran a third arm with every decision frozen to its default, which has zero decision variance at all. One hundred tournaments per arm, same seeds, same squads.

Results

Arm	Runs	Distinct champions	Entropy (bits)	Most titles	Goals/match
LLM agents	100	25	4.30	Spain, 12%	2.69
Random agents	100	24	4.16	Spain, 13%	2.57
Default decisions	100	27	4.48	Argentina, 11%	2.58

A chi-square test cannot tell the LLM and random champion tables apart ($p = 0.28$). The decisive check is the split-half baseline: two halves of the same arm differ from each other by a total variation distance of 0.36 to 0.42, while the LLM arm and the random arm differ by only 0.28. The difference between intelligent agents and coin-flippers is smaller than the difference between two samples of the same coin-flipper.

Title odds with the agents' brains removed — 100 tournaments per arm



Title odds with the agents' brains removed. Three arms, 100 tournaments each.

Verdict: the flat champion field is sampling noise, not emergent tactics. In this architecture, agents change the story of a match, not who wins the Cup.

I want to be precise about what this does and does not say. The agents do change real things: who scores, which chances die at the keeper's feet, who gets substituted at half time, the texture and narrative of every individual match. What they do not change, at this sample size and with my safety clamps, is the championship distribution. The spread comes from the calibrated Poisson goal process plus the brutality of single-elimination knockouts. The plus or minus fifteen percent clamps that saved my calibration also capped the agents' influence on outcomes. If I wanted agents to move outcomes, I would have to give them bigger levers, and then I would owe the calibration check all over again.

This is a negative result and I am documenting it anyway, because it is the most useful finding in the project. It took one afternoon and it changed the headline.

7. The prediction engine and the final answer

The ablation has a practical consequence: if agent decisions do not move championship outcomes, then for actually predicting the winner the LLM layer is the wrong tool. So I built a third engine in a separate project folder with one job only: the sharpest win percentages I can get from public signals. No LLM anywhere in it.

The match model

Each match is resolved from the Elo rating gap between the two teams. I convert the gap into two Poisson goal means, solved numerically so that the resulting win, draw, and loss probabilities reproduce the Elo win expectancy exactly, with total goals fixed at the real World Cup average of 2.64. Draws fall out naturally, knockouts get extra time at a reduced scoring rate and penalties with a small Elo tilt, and the three host nations get a standard home-advantage bump of 80 Elo points. The bracket logic is reused from the validated engine, including the real results already played this week.

Validation first

Before producing a single 2026 number I backtested the model on the four previous World Cups, 2,000 simulations each:

Year	Actual champion	Model probability	Model rank of actual	Model top three
2010	Spain	22.9%	1	Spain 23, Brazil 23, Netherlands 14
2014	Germany	17.2%	3	Brazil 29, Spain 20, Germany 17
2018	France	7.0%	4	Brazil 34, Germany 19, Spain 15
2022	Argentina	26.9%	2	Brazil 31, Argentina 27, Netherlands 8

The actual champion lands in the model's top four every time, carrying 7 to 27 percent probability. That is comparable to bookmaker priors. The backtest also exposes the model's known bias: Elo overrates Brazil in every single year, because Elo measures long-run strength and is slow to price form.

The market blend

With the model validated, I ran 20,000 Monte Carlo tournaments of the 2026 bracket. It takes about eight seconds, which is a useful reminder that the agentic engine spent GPU-hours answering a question this engine answers in the time it takes to read this sentence. The raw model says Spain 38.9 percent, which is too confident for the same reason it loved Brazil in the backtests. Bookmakers price Spain at about 15.5 percent after removing their margin, because the market knows things Elo cannot: France's friendly defeat to Ivory Coast last week, Portugal money arriving, injury news. So the final number is a geometric blend of the de-vigged market and the model, equal weights. The model contributes the bracket math; the market contributes the news.

Country	Final win %	Elo model %	Market %
Spain	27.3	38.9	15.5
Argentina	14.2	21.0	7.8
France	12.8	9.0	14.8
England	11.3	10.2	10.0
Brazil	7.0	4.8	8.1
Portugal	6.5	3.6	9.5
Germany	3.9	2.1	5.7
Netherlands	3.7	2.7	4.1

Final prediction, June 12, 2026: Spain 27 percent, Argentina 14, France 13, England 11. Spain is the most likely champion in every method I built, but a 73 percent chance the winner is somebody else is the honest headline.

It is worth noting where my model and the market disagree most. The model is much hotter on Spain and Argentina than the market, and cooler on France and Portugal. Where a calibrated model and an efficient market disagree, the truth usually sits in between, which is exactly what the blend assumes.

8. What I learned

1. **Calibrate before you believe anything.** My first agentic engine produced exciting, fluent, completely wrong football: 3.61 goals per match. Without a calibration target from real data I would have shipped it.
2. **An LLM has a personality, and an unconstrained simulation will express it.** The 95 percent passing-goal rate was qwen's pass bias amplified by my reward structure. The simulation measured the model, not the sport.
3. **The clamp that saves your physics caps your agents.** Bounded steering kept the engine honest and simultaneously guaranteed that agent intelligence could not move championship outcomes much. That tradeoff was invisible until the ablation made it explicit.
4. **Run the ablation. It is cheap and it is decisive.** One hundred coin-flipper tournaments ran locally in minutes and settled a question I had been hand-waving about. The difference between agents and randomness was smaller than the difference between two samples of randomness.
5. **LLM agents are a narrative engine, not a prediction engine.** They generate matches that feel like real football, including a tournament that spontaneously re-created 2018's emotional arc. For stories, they are remarkable. For probabilities, an eight-second Elo Monte Carlo beat GPU-hours of agents.
6. **Blend models with markets.** The model knows the bracket; the market knows the news. Each one corrects the other's characteristic failure.
7. **Negative results are results.** "My agents do not change outcomes, and here is the experiment that proves it" is more valuable, and more credible, than another demo claiming emergence.
8. **Operational scars are part of the study.** Cookie accumulation breaking an API after thousands of calls, carriage returns in headers, duplicate names from a PDF, an output prefix collision that nearly cost a batch of runs. Every one of these is now a guard in the code.

9. Limitations and next steps

- **Sample sizes.** One hundred tournaments per agentic arm bounds the effects I can detect. If agents move title odds by two or three points, I would need thousands of runs to see it. The ablation says any effect is small, not that it is exactly zero.
- **Stateless agents.** My agents decide each moment from a short prompt with no memory across matches. Persistent personas with form, confidence, and accumulated fatigue might behave differently.
- **Tight clamps by design.** Wider tactical bounds with a fresh calibration pass is the obvious next experiment: give managers plus or minus thirty-five percent and recalibrate the base scoring rate, then re-run the ablation and see if intelligence starts to matter.

- **Elo's form blindness.** The prediction engine inherits Elo's slow reaction to news. The market blend patches this, but a model with explicit form and injury inputs would be better.
- **Static market snapshot.** The blend uses one day's odds. Odds will move all tournament; the pipeline reruns in seconds, so refreshing is cheap.
- **One model family.** All agentic results are qwen2.5 14B. A different model would have different biases, and the v1 failure suggests those biases would leak into any unconstrained simulation.

10. Artifact index

Artifact	What it is
<code>worldcup_2026_sim.py</code>	Core engine: features, players, Poisson match model, 48-team bracket, backtest support
<code>run_2026.py</code> / <code>out_2026_probs.csv</code>	Statistical engine, 100 tournaments, results
<code>agentic_wc2026.py</code>	Agentic engine v2 with manager, player, and keeper agents, plus the random-agents ablation mode
<code>agentic_champions_pooled.csv</code>	Champions across the 100 agentic tournaments
<code>ablation_analysis.py</code> / <code>ablation_results.md</code>	Three-arm ablation, statistics, and verdict
<code>../fifa_predict/elo_predict.py</code>	Elo Monte Carlo prediction engine with 2010-2022 backtest
<code>../fifa_predict/market_blend.py</code> / <code>final_2026_prediction.csv</code>	Market blend and the final win percentages
<code>viz_*.png</code>	All figures in this document
<code>wc2026_apple_deck.pptx</code> / <code>VIDEO_SCRIPT_V2.md</code>	Presentation deck and video script
<code>final_report_100.html</code>	Interactive two-engine report

Written June 12, 2026, the second day of the tournament. One of these brackets gets to be right in July.